

YANGU Social Media Engine

Full Technical Audit & Architecture Report

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Version: 1.0
Prepared for: YANGU Platform

1. Executive Summary

The YANGU Social Media Engine is a full-stack module designed to power multi-platform social media management within the YANGU ecosystem. This audit documents the complete architecture, database schema, integration layer, services, hooks, components, and current implementation status as of March 29, 2026.

The module uses a six-layer architecture: UI → State Management → Domain Services → Data Access → Provider Integration → AI Engine. Outstand serves as the primary third-party provider for OAuth-based account connection, content publishing, scheduling, and analytics.

2. Architecture Overview

2.1 Six-Layer Architecture

Layer	Responsibility	Status
UI Layer	Pages, components, shared UI primitives	☐ Built
State Layer	React Query hooks, query keys, cache	☐ Built
Domain Layer	Post lifecycle, analytics, library services	☐ Built
Data Layer	Supabase client, DB queries, RLS	☐ Built

Integration Layer	Outstand adapter, edge proxy	□ Live-wired
AI Engine	7-layer generation pipeline	△ Scaffolding

2.2 Internal Navigation Structure

The Social Media module uses a horizontal top navigator with grouped sections:

← Dashboard | + Create Post | Home | CONTENT (Posts, Calendar, Analytics) | STRATEGY (Library, Topics, AI Profile, Workspace)

3. Database Schema

3.1 Tables (13 Total)

Table	Purpose	RLS Policy
social_workspaces	User workspace config, goals, frequency	Users manage own
social_connected_accounts	OAuth-linked social accounts	Users manage own
social_brand_profiles	Brand voice, tone, rules, keywords	Users manage own
social_onboarding	Onboarding step tracking	Users manage own
social_posts	Post content, status, scheduling	△ Admin-only (ISSUE)
social_post_targets	Per-account publish targets	Via post ownership join
social_post_variants	Platform-adapted caption variants	Via post ownership join
social_publish_events	Lifecycle event log	Read via target → post join
social_analytics_snapshots	Analytics data points	Users read own
social_topics	Content topic management	Users manage own
social_topic_categories	Topic category grouping	Users manage own
social_library_items	Media/content library assets	Users manage own
social_connections	Legacy admin connections	Admin-only (legacy)

3.2 Critical RLS Issue

△ **social_posts table has ADMIN-ONLY RLS policy.**

Regular authenticated users cannot create, read, update, or delete their own posts. This blocks the entire post creation, scheduling, and publishing flow for non-admin users. A user-level policy must be added.

4. Backend / Edge Functions

4.1 outstand-proxy

Single edge function handling all Outstand API communication. Base URL:

<https://api.outstand.so/v1>

Action	Method	Outstand Endpoint	Status
health	GET	/accounts (lightweight)	☐ Live
get_connect_url	POST	/auth/connect	☐ Live
oauth_callback	POST	/auth/callback + DB persist	☐ Live
list_accounts	GET	/accounts	☐ Live
refresh_account	GET	/accounts/:id	☐ Live
disconnect_account	POST	/accounts/:id/disconnect	☐ Live
publish_post	POST	/posts	☐ Live
schedule_post	POST	/posts/schedule	☐ Live
fetch_analytics	GET	/analytics	☐ Live
webhook	N/A	Event intake + DB logging	☐ Live

4.2 Secrets Configuration

Secret	Location	Status
OUTSTAND_API_KEY	Project secrets (backend only)	☐ Added
SUPABASE_URL	Auto-provided	☐ Available
SUPABASE_SERVICE_ROLE_KEY	Auto-provided	☐ Available

4.3 URLs to Register with Outstand

Purpose	URL
OAuth Callback	https://xcipuyvcwfytljsryhvs.supabase.co/functions/v1/outstand-proxy
Webhook Intake	https://xcipuyvcwfytljsryhvs.supabase.co/functions/v1/outstand-proxy
Frontend Redirect	/dashboard/social-media/callback

5. Domain Services

Service	File	Functions	Status
Provider Interface	providerInterface.ts	Adapter contract + ProviderRegistry	☐ Complete
Outstand Adapter	outstandAdapter.ts	All 10 provider methods via proxy	☐ Live
Post Lifecycle	postLifecycleService.ts	Create, schedule, publish, archive, delete, duplicate, retry	☐ Wired
Analytics	analyticsService.ts	Snapshots, workspace summary, post analytics	☐ Wired

Library	libraryService.ts	List, import, upload, website import	⚠ Scaffolding
AI Engine	aiEngine.ts	7 sub-engines (see section 7)	⚠ Scaffolding

5.1 Post Lifecycle State Machine

Valid transitions:

- draft → ready, scheduled, archived
- ready → draft, scheduled, publishing, archived
- scheduled → draft, ready, publishing, archived
- publishing → published, failed
- published → archived
- failed → draft, ready, scheduled, archived
- archived → draft

6. React Hooks

Hook	Purpose	Status
useConnectedAccounts	List/connect/disconnect accounts via Outstand	☐ Live
useSocialPosts	CRUD + lifecycle mutations for posts	⚠ Wired (blocked by RLS)
useSocialWorkspace	Create/read workspace	☐ Wired
useSocialHomeSummary	Aggregated dashboard counts	☐ Wired
useSocialBrandProfile	Brand voice configuration	☐ Wired
useSocialAnalytics	Analytics data fetching	☐ Wired
useSocialCalendar	Calendar view data	☐ Wired
useSocialTopics	Topic management	☐ Wired
useSocialLibrary	Library items	⚠ Scaffolding (returns empty)
useSocialOnboarding	Onboarding state (localStorage)	☐ Working
queryKeys	Full query key factory	☐ Complete

7. AI Engine Architecture

The AI engine is designed as a 7-layer pipeline. All layers are currently scaffolding with interface contracts but return placeholder data.

Layer	Engine	Purpose	Status
1	Brand Context Builder	Extract business intelligence from website/description	⚠ Scaffold
2	AI Profile Rules Engine	Convert brand profile to generation rules	☐ Working (local logic)

3	Topic Engine	Generate topic suggestions via AI	⚠ Scaffold
4	Caption Generator	Create captions with AI models	⚠ Scaffold
5	Post Variant Generator	Generate full post variations from ideas	⚠ Scaffold
6	Media Prompt Builder	Generate image prompts from captions	⚠ Scaffold
7	Platform Adaptation Engine	Adapt content per platform limits	❑ Working (local logic)

8. UI Components & Pages

8.1 Pages

Page	Route	Status
Layout	/dashboard/social-media/*	❑ Working (top nav)
Home	/dashboard/social-media	❑ Working (progress + links)
Posts	/dashboard/social-media/posts	▤ Shell only
Calendar	/dashboard/social-media/calendar	▤ Shell only
Analytics	/dashboard/social-media/analytics	▤ Shell only
Library	/dashboard/social-media/library	▤ Shell only
Topics	/dashboard/social-media/topics	▤ Shell only
AI Profile	/dashboard/social-media/ai-profile	▤ Shell only
Workspace	/dashboard/social-media/workspace	▤ Shell only (static list)

8.2 Components

Component	Purpose
SocialOnboardingFlow	4-step guided onboarding modal
SocialEmptyState	Reusable empty state placeholder
SocialFilterBar	Filter controls for lists
SocialProgressChecklist	Progress tracking checklist
SocialProviderRow	Provider connection row item
SocialQuickActionCard	Quick action shortcut card
SocialSectionHeader	Section title header
SocialStatusPill	Status badge component

8.3 Onboarding Flow (4 Steps)

- Step 1: Create Workspace — Website URL or business description input
- Step 2: Content Plan — Posting goals selection + frequency

- Step 3: Connect Socials — Provider list with real OAuth connect buttons
- Step 4: Instagram Help — Pre-auth guidance modal for Instagram/Story

9. Type System

All types defined in `src/types/socialMedia.ts` (444 lines). Key type groups:

Category	Types
Enums	SocialProvider (11), PostStatus (7), PostSourceType (4), PostContentType (5), AccountConnectionStatus (5), OnboardingStep (4), PublishEventType (10), LibraryItemType (12), LibraryItemStatus (4), TopicSourceType (3), ApprovalStatus (4)
Core Entities	SocialWorkspace, SocialConnectedAccount, SocialBrandProfile, SocialTopicCategory, SocialTopic, SocialLibraryItem, SocialPost, SocialPostVariant, SocialPostTarget, SocialPublishEvent, SocialAnalyticsSnapshot, SocialOnboardingState
Aggregates	SocialHomeSummary, PostCounts, AnalyticsDateRange, AnalyticsSummary
Form/Input	CreatePostInput, UpdatePostInput, CreateWorkspaceInput, UpdateBrandProfileInput, ImportLibraryInput
Provider	ProviderConnectResult, ProviderCallbackResult, ProviderPublishResult, ProviderAnalyticsResult, ProviderError

10. What Needs to Be Done Next

10.1 Critical Fixes

- 1. Fix `social_posts` RLS** — Add user-level policy so non-admin users can CRUD their own posts
- 2. Add `social_publish_events` INSERT policy** — Edge function uses service role but client-side inserts will fail
- 3. Build OAuth callback route** — `/dashboard/social-media/callback` page does not exist yet

10.2 Onboarding Improvements

- 4. Persist onboarding state to `social_onboarding` table instead of `localStorage`
- 5. Instagram help modal Continue button should initiate real Outstand connect

10.3 UI Sections to Build

- 6. Posts page — List, create, edit, delete posts with status filters
- 7. Calendar page — Visual calendar with scheduled/published posts
- 8. Analytics page — Charts and metrics from snapshot data
- 9. Library page — Upload, import, browse media/content assets
- 10. Topics page — Manage topic categories and individual topics
- 11. AI Profile page — Brand voice and tone configuration form
- 12. Workspace page — Real connected accounts list, settings management

10.4 AI Engine Activation

- 13. Wire Brand Context Builder to Lovable AI (Gemini/GPT)
- 14. Wire Caption Generator to produce real captions
- 15. Wire Topic Engine for AI-powered topic suggestions
- 16. Wire Media Prompt Builder for image generation prompts
- 17. Wire Post Variant Generator for multi-platform content

10.5 Integration Completion

- 18. Register callback & webhook URLs with Outstand
- 19. Implement webhook signature verification
- 20. End-to-end test: connect account → create post → publish
- 21. Wire library service to real social_library_items table queries

10.6 Summary Status

Area	Status
Database (13 tables + RLS)	⚠ Complete (1 RLS fix needed)
Edge Function (outstand-proxy)	❑ Live-wired (10 actions)
Provider Adapter	❑ Live-wired
Domain Services (6)	❑ 4 wired, 2 scaffolded
React Hooks (11)	❑ 9 wired, 2 scaffolded
Type System	❑ Complete (444 lines)
UI Layout + Navigation	❑ Working
Onboarding Flow	❑ Working (needs DB persist)
Home Page	❑ Working
Section Pages (7)	▣ Shell only
AI Engine (7 layers)	⚠ Scaffolded
Shared Components (7)	❑ Built